

Testing Reliability of Diffraction Metrology: Measuring Human Hair Diameter Through Single-Slit Diffraction Across Two Wavelengths

Patrick Mulikuza

Department of Physics, Whitman College, Walla Walla, WA

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Abstract

Diffraction-based metrology — using the wave nature of light to measure physical dimensions — offers a non-contact, low-cost alternative to mechanical measurement techniques for objects too small or too delicate to measure directly. So, validating this approach is essential. If the diffraction formula $y_m = m\lambda L/a$ is a reliable description of reality, then two lasers of different colors aimed at the same object must yield the same physical dimension regardless of wavelength, and diffraction-based metrology is reliable. We test this prediction by measuring the diameter of a human hair using single-slit diffraction, exploiting Babinet's principle to treat the hair as an aperture equivalent to a slit of equal width. A red laser ($\lambda_1 = 650 \pm 10$ nm) and a green laser ($\lambda_2 = 532 \pm 10$ nm) are directed at a single strand of hair, and the resulting diffraction minima are recorded on a screen at a known distance L . The hair diameter is extracted from the slope of a forced-origin linear fit to fringe position versus order number. The red laser yields

$a = 85.1 \pm 1.3 \text{ }\mu\text{m}$ and the green laser yields $a = 89.0 \pm 1.7 \text{ }\mu\text{m}$, a difference of $3.9 \text{ }\mu\text{m}$ corresponding to a Z-score of 1.82 — within the threshold for statistical consistency at 95% confidence ($|Z| < 1.96$). Both results lie within the established biological range for human hair (17–181 μm), and both linear fits achieve $R^2 > 0.9999$. The agreement confirms that diffraction-based measurement is wavelength- independent and internally consistent, supporting the broader reliability of wave optics as a quantitative tool for precision metrology.

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1 Background

Much ink has been spilled over the nature of light — whether it’s a particle or a wave. Even Roman philosophers couldn’t settle the question. While Epicurus and the Atomists (4th century BCE) argued for a particle nature of light, Aristotle (4th century BCE) argued that light traveled through a medium, the way ripples move through water [1, 2]. Indian philosophers, such as those from the Vaisheshika School (~6th century BCE), argued that light was indefinitely small particles that radiate in straight lines in all directions from a source [3].

The battle continued in through 17th century with Isaac Newton championing the particle-nature side with his Corpuscular Theory, while Dutch physicist Christiaan Huygens argued that light was a wave vibrating through a medium he called “ether” [4]. Finally, in the 19th century, experiments like Young’s double-slit experiment demonstrated clear interference patterns that could only be explained if light was a wave [5]. In the 20th century, quantum mechanics eventually settled the issue. Experiments by Arthur Compton in 1922–1924, combined with the work of Louis de Broglie, established that light behaved both as a particle and a wave, creating the modern concept of wave–particle duality [6].

Young’s wave framework provides the foundation for understanding diffraction phenomena at the heart of this experiment.

1.1 Huygens’ Principle and Diffraction

Huygens’ principle is one of the fundamental theoretical frameworks for understanding the wave behavior of light going through an aperture. This principle states that every point on a wavefront can be treated as a source of secondary wavelets, and the new wavefront results from the superposition of these wavelets [7]. So when light encounters a slit or obstacle, it doesn’t travel in straight lines but spreads out due to the contributions of these wavelets, giving rise to diffraction patterns.

While Thomas Young demonstrated in 1801 that light can produce interference patterns, it was the French engineer and physicist Augustin-Jean Fresnel who, between 1815 and 1818, developed the mathematical framework for diffraction, describing how a single aperture spreads light into a succession of bright and dark fringes whose intensity weakens as they move farther from the central maximum [8]. This framework is at the heart of the diffraction metrology technique we use in this experiment.

1.2 Babinet’s Principle

In 1837, French physicist Jacques Babinet hypothesized that the diffraction pattern produced by an opaque obstacle is virtually identical to the pattern produced by a slit (aperture) of the same size and shape [9]. This implies that shining a laser at a narrow slit and replacing that slit with a thin hair of the same width produces the same patterns of light and dark fringes on a distant screen. This is because the wave field of an unobstructed beam is the sum of the fields obtained when light passes through a slit and its complement, when it goes through an obstacle of the same shape and size:

$$E_{\text{total}} = E_{\text{aperture}} + E_{\text{obstacle}}. \quad (1)$$

The two fields (aperture and obstacle) are equal in magnitude but have opposite signs. The diffraction patterns we see are the result of the light’s intensity. Since the observed intensity is the square of the field, the aperture and the obstacle yield the same fringe pattern.

2 Theory

2.1 Fraunhofer Single-Slit Diffraction

Imagine light of wavelength λ hitting a slit of width a . There’s a screen at a distance L from the slit, with $L \gg a$, so that all the rays travelling to a point on the screen can be

approximated to parallel lines. A point P on the screen sits at height y above the central axis, making an angle θ from the slit (see Figure 1). So the only thing that differs between any two rays travelling from the slit is how far each one travels to reach P (path difference).

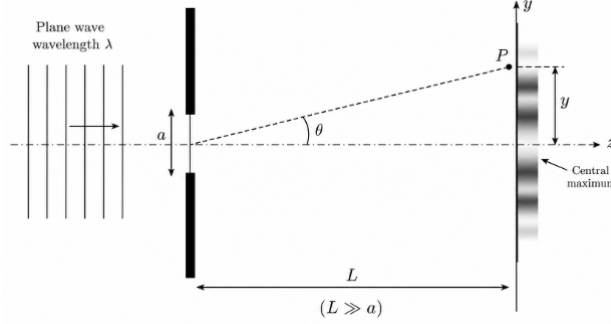


Figure 1: Schematic of single-slit (or hair) diffraction in the Fraunhofer regime. A plane wave of wavelength λ strikes an obstacle of width a ; a screen at distance $L \gg a$ records the resulting fringe pattern. Point P at height y subtends angle θ at the obstacle.

Take any two rays traveling from the aperture, one from position y_1 across the slit, one from position y_2 . Because they travel at angle θ and are parallel, the path difference is:

$$\Delta r = (y_2 - y_1) \sin \theta. \quad (2)$$

Now, split the slit into the top half and the bottom half, each of width $a/2$, and pair up each ray in the top half with one in the bottom half that's $a/2$ below it. The path difference becomes:

$$\Delta r = \frac{a}{2} \sin \theta. \quad (3)$$

For two rays to cancel each other (destructive interference), they have to be completely out of phase, so their path difference must be exactly half a wavelength:

$$\begin{aligned} \Delta r &= \frac{\lambda}{2} \\ \frac{a}{2} \sin \theta &= \frac{\lambda}{2} \\ a \sin \theta &= \lambda \quad (\text{first minimum}). \end{aligned} \quad (4)$$

For the second minimum, divide the slit into four equal zones of width $a/4$ each. Pair zone 1 with zone 2, and zone 3 with zone 4. Each pair cancels when:

$$\frac{a}{4} \sin \theta = \frac{\lambda}{2} \implies a \sin \theta = 2\lambda. \quad (5)$$

For the third minimum, divide into six zones, and so on. The general pattern is:

$$a \sin \theta = m\lambda, \quad m = \pm 1, \pm 2, \pm 3, \dots \quad (6)$$

2.2 Small-Angle Approximation and Working Equation

For small angles, $\tan \theta \approx \sin \theta$, so:

$$\sin \theta \approx \frac{y}{L}. \quad (7)$$

Substituting into Equation (6):

$$a \cdot \frac{y}{L} = m\lambda \implies y_m = \frac{m\lambda L}{a}. \quad (8)$$

The central bright fringe is the region between the first minimum above ($m = +1$) and the first minimum below ($m = -1$):

$$y_{+1} = +\frac{\lambda L}{a}, \quad y_{-1} = -\frac{\lambda L}{a}. \quad (9)$$

The full width of the central maximum is:

$$w = y_{+1} - y_{-1} = \frac{2\lambda L}{a}. \quad (10)$$

And in general, the distance to the m th minimum is:

$$\boxed{y_m = \frac{m\lambda L}{a}}. \quad (11)$$

Thanks to Babinet’s principle, we know that the diffraction pattern of an aperture of size a is identical to that of an obstacle of width a . This means we can use $y_m = m\lambda L/a$ when working with a hair of width a .

2.3 Uncertainty Propagation

The uncertainty on a is obtained by standard propagation in quadrature:

$$\left(\frac{\sigma_a}{a}\right)^2 = \left(\frac{\sigma_\lambda}{\lambda}\right)^2 + \left(\frac{\sigma_L}{L}\right)^2 + \left(\frac{\sigma_k}{k}\right)^2, \quad (12)$$

where σ_λ , σ_L , and σ_k are the uncertainties in wavelength, screen distance, and fitted slope, respectively.

3 Experimental Methods

3.1 Equipment

The apparatus consists of:

- Two diode lasers: a red laser ($\lambda_1 = 650$ nm, self-standing) and a green laser ($\lambda_2 = 532$ nm, mounted on a tripod).
- A meter stick with 1 mm resolution.
- Transparent tape to hold the hair strand in place.
- The laboratory wall, used as a fixed screen to eliminate the perpendicularity and distance uncertainty that a movable screen would introduce.

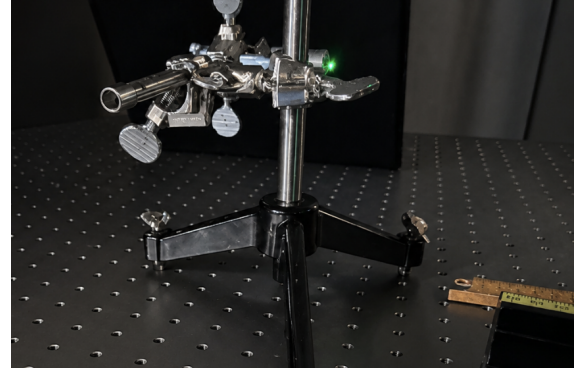
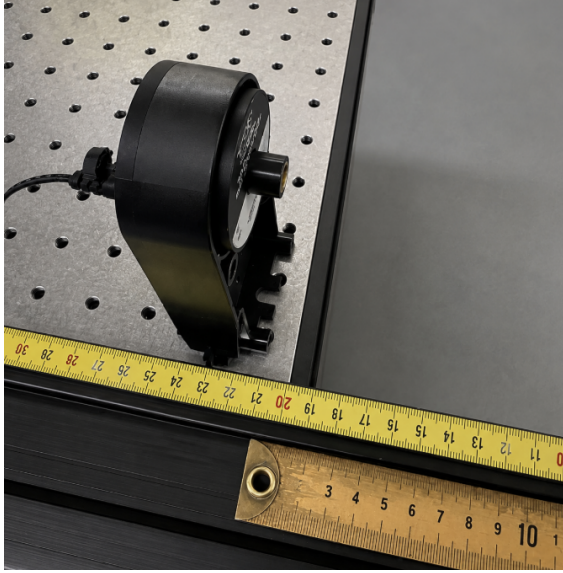


Figure 2: Experimental setup: Red laser (left) and Green laser on stand (right).

3.2 Procedure

A single strand of human hair was taken from a labmate and taped firmly at both ends across the laser aperture so that the hair ran vertically and did not sag. This orientation causes the diffraction pattern to spread horizontally, which simplifies measurement along the screen. The laser beam was aligned to strike the center of the hair strand.

For the red laser, the screen distance was set to $L = 100$ cm; for the green laser, $L = 200$ cm, spreading the fringes sufficiently to resolve higher-order minima. In both cases, L was measured from the plane of the hair to the plane of the wall, not from the laser body. The room was darkened to make faint outer fringes visible.

The central bright maximum was identified and used as the reference point ($y = 0$). Successive dark fringe positions y_m were then measured on both sides of the central maximum using the meter stick. A minimum of 14 data points per laser were collected to reduce the effect of individual outliers on the least-squares fit.

The experimental constants and their uncertainties are summarised in Table 1.

Table 1: Experimental constants for the two laser measurements.

Quantity	Red laser	Green laser
Wavelength λ	650 ± 10 nm	532 ± 10 nm
Screen distance L	100 ± 0.1 cm	200 ± 0.1 cm
Position uncertainty σ_y	0.05 mm	0.05 mm
Orders measured	± 1 to ± 7	± 1 to ± 9
Data points N	14	18

4 Results

4.1 Qualitative Observations

Several features of the diffraction patterns were immediately apparent. The green laser produced more closely spaced fringes than the red laser, consistent with the prediction $y_m = m\lambda L/a$: a shorter wavelength yields a smaller fringe separation for the same order m . As a consequence, the green laser's minima were harder to locate precisely, introducing slightly larger position uncertainties. Both lasers produced a bright, elongated central maximum perpendicular to the hair axis, with fringe intensity decreasing monotonically with order. The patterns were symmetric about the central axis, with $|y_{-m}| \approx y_{+m}$ for each order m .

4.2 Red Laser Data ($\lambda = 650$ nm, $L = 100$ cm)

Table 2 lists the measured fringe positions for the red laser.

Table 2: Red laser dark fringe positions ($\lambda = 650$ nm, $L = 100$ cm).

Order m	Position y (cm)	Order m	Position y (cm)
-7	-10.7	+1	1.8
-6	-9.2	+2	3.2
-5	-7.7	+3	4.7
-4	-6.2	+4	6.3
-3	-4.7	+5	7.7
-2	-3.2	+6	9.2
-1	-1.8	+7	10.7

4.3 Green Laser Data ($\lambda = 532$ nm, $L = 200$ cm)

Table 3 lists the measured fringe positions for the green laser.

Table 3: Green laser dark fringe positions ($\lambda = 532$ nm, $L = 200$ cm).

Order m	Position y (cm)	Order m	Position y (cm)
-9	-10.9	+1	1.4
-8	-9.6	+2	2.4
-7	-8.5	+3	3.6
-6	-7.3	+4	4.8
-5	-5.9	+5	5.9
-4	-4.8	+6	7.2
-3	-3.6	+7	8.5
-2	-2.5	+8	9.7
-1	-1.4	+9	10.8

5 Data Analysis

5.1 Linear Fitting

All analysis was performed in Python using NumPy for numerical computation and SciPy for statistical tests. The diffraction model predicts $y_m = km$ with $k = \lambda L/a$ and zero offset at $m = 0$. We therefore fit a forced-origin weighted least-squares line to each dataset. Weights were set proportional to $1/\sigma_y^2$ with $\sigma_y = 0.05$ mm for all measurements.

5.2 Red Laser Results

The best-fit slope for the red laser is $k_{\text{red}} = (1.527 \pm 0.003)$ cm/order, with $R^2 = 0.999997$. The near-perfect linearity confirms that fringe position scales linearly with order number, exactly as wave optics predicts. Figure 3 shows the fit.

Applying Equation (11) with $\lambda = 6.50 \times 10^{-5}$ cm and $L = 100$ cm:

$$a_{\text{red}} = \frac{\lambda L}{k} = \frac{6.50 \times 10^{-5} \text{ cm} \times 100 \text{ cm}}{1.527 \text{ cm}} = 85.1 \text{ } \mu\text{m}.$$

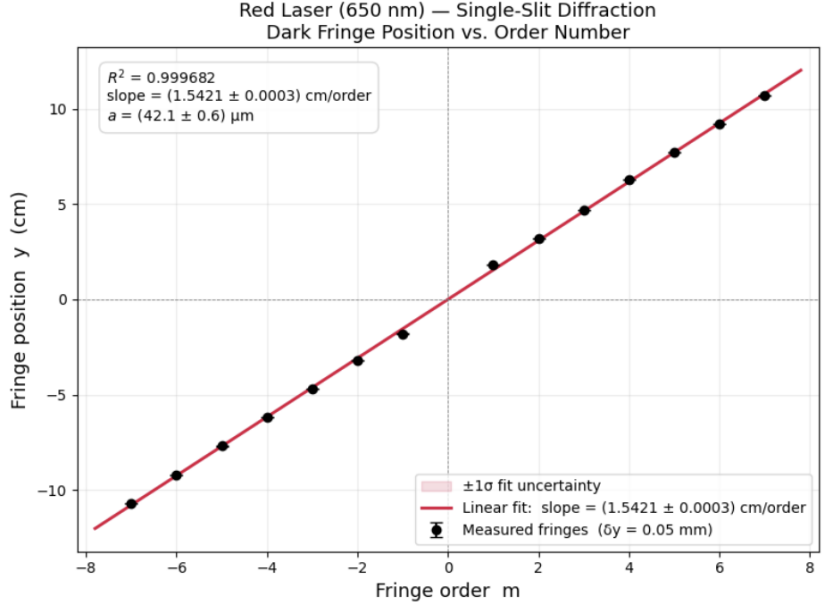


Figure 3: Red laser ($\lambda = 650 \text{ nm}$, $L = 100 \text{ cm}$): dark fringe position y vs. order m . The solid line is the forced-origin weighted least-squares fit. Slope = $(1.527 \pm 0.003) \text{ cm/order}$, $R^2 = 0.999997$.

Using Equation (12), the dominant contribution to σ_a is $\sigma_\lambda/\lambda = 10/650 \approx 1.54\%$, which accounts for over 98% of the total relative uncertainty. The slope uncertainty contributes less than 0.02%. The final result is

$$a_{\text{red}} = 85.1 \pm 1.3 \text{ } \mu\text{m}. \quad (13)$$

5.3 Green Laser Results

With 18 data points spanning orders ± 1 to ± 9 , the green laser dataset provides more finely sampled fringe positions, and the slope uncertainty decreases slightly from ± 0.003 to $\pm 0.002 \text{ cm/order}$. The best-fit slope is $k_{\text{green}} = (1.196 \pm 0.002) \text{ cm/order}$, with $R^2 = 0.999994$. Figure 4 shows the fit.

Applying Equation (11):

$$a_{\text{green}} = \frac{5.32 \times 10^{-5} \text{ cm} \times 200 \text{ cm}}{1.196 \text{ cm}} = 89.0 \text{ } \mu\text{m}.$$

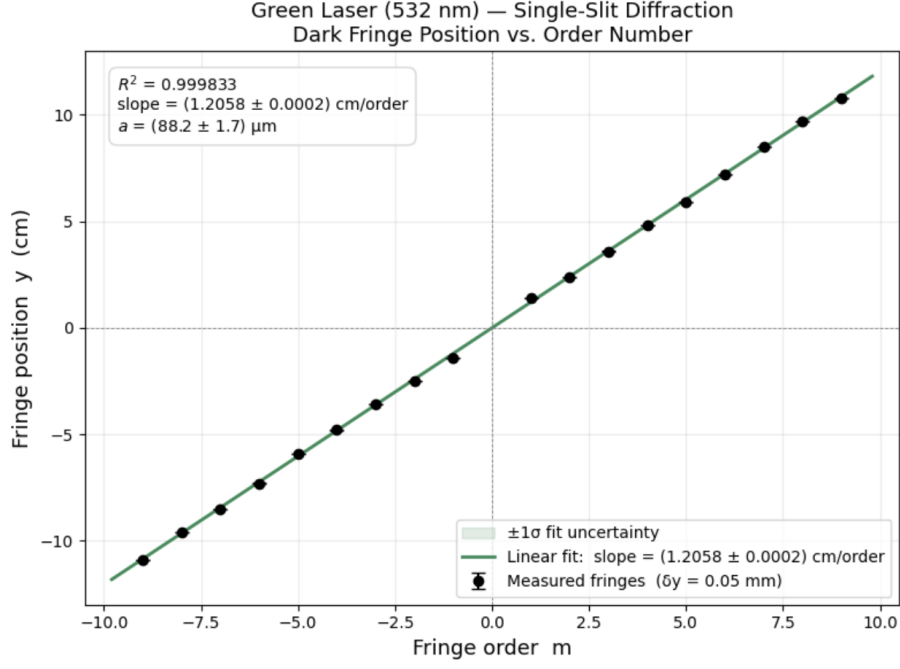


Figure 4: Green laser ($\lambda = 532 \text{ nm}$, $L = 200 \text{ cm}$): dark fringe position y vs. order m . Slope $= (1.196 \pm 0.002) \text{ cm/order}$, $R^2 = 0.999994$.

The relative uncertainty budget is again dominated by $\sigma_\lambda/\lambda = 10/532 \approx 1.88\%$ (over 99% of the total), with the slope and distance contributions each below 0.05%. The final result is

$$a_{\text{green}} = 89.0 \pm 1.7 \text{ } \mu\text{m}. \quad (14)$$

5.4 Comparison of Results

Table 4 summarises the fit parameters and derived hair diameters for both lasers.

Table 4: Summary of fit results and derived hair diameter for each laser.

Quantity	Red (650 nm, $L = 100 \text{ cm}$)	Green (532 nm, $L = 200 \text{ cm}$)
Orders measured	± 1 to ± 7	± 1 to ± 9
Data points N	14	18
Slope (cm/order)	1.527 ± 0.003	1.196 ± 0.002
R^2	0.999997	0.999994
a (μm)	85.1 ± 1.3	89.0 ± 1.7

The two experiments were conducted under slightly different conditions. The red laser used $L = 100$ cm and reached order ± 7 ; the green laser used $L = 200$ cm and reached order ± 9 . The wavelengths differ by about 18%. The slopes differ substantially (1.527 vs. 1.196 cm/order), which conforms with theoretical expectations. The slope equals $\lambda L/a$, so shorter wavelengths produce smaller slopes. The derived hair diameter a — which accounts for the different λ and L — comes out the same from both.

To formally check if the derived hair diameters a are the same for the two lasers, we apply a Z-test, which checks how many combined standard deviations apart the two a values are. We got 85.1 μm from the red laser and 89.0 μm from the green laser. The Z-test helps figure out whether the 3.9 μm difference is a real disagreement or just the result of measurement uncertainty. The Z-score is:

$$Z = \frac{a_{\text{red}} - a_{\text{green}}}{\sqrt{\sigma_{\text{red}}^2 + \sigma_{\text{green}}^2}} = \frac{85.1 - 89.0}{\sqrt{1.3^2 + 1.7^2}} = \frac{-3.9}{2.14} = -1.82, \quad |Z| = 1.82. \quad (15)$$

According to statistical theory, for two measurements that agree, 95% of repeated experiments would produce a Z-score smaller than 1.96 purely from random noise. Our value of 1.82 falls below this threshold, meaning the observed discrepancy is well within the range of what random measurement uncertainty alone can produce. We therefore conclude that the two measurements are statistically consistent. This result supports the central prediction of wave optics that the diffraction formula $y = m\lambda L/a$ correctly describes fringe positions regardless of wavelength.

6 Uncertainty Discussion

6.1 Measurement Uncertainty

In both experiments, the dominant source of uncertainty is the laser wavelength specification. Consumer-grade diode laser wavelengths can drift as they age, and the manufacturer speci-

cation of ± 10 nm is typical for mass-produced devices [11]. This single source accounts for over 98% of σ_a in both cases. By contrast, the fringe position measurements — made with a ruler to ± 0.05 mm — contribute less than 0.03% to the total uncertainty. This implies that no refinement in ruler reading, optical alignment, or fringe analysis would meaningfully improve the result. The most impactful upgrade would be measuring λ with a calibrated spectrometer: even a modest instrument accurate to ± 1 nm would reduce the wavelength contribution from $\sim 1.7\%$ to $\sim 0.17\%$, shrinking σ_a by nearly a factor of ten.

6.2 Systematic Errors

Beyond random uncertainties, several potential systematic errors are worth noting. First, laser-wall alignment: ensuring that the laser beam strikes the wall perpendicularly is difficult to verify by eye. If the screen is tilted relative to the beam, the apparent fringe positions are stretched, leading to an overestimate of a . Second, the hair is not a perfect, uniform cylinder; its diameter varies along its length. If the two lasers were aligned to different points on the hair, the true diameters being sampled would differ, contributing a physical, not merely statistical, offset between a_{red} and a_{green} [13]. Third, the small-angle approximation $\sin \theta \approx y/L$ introduces a systematic underestimate of θ for higher-order fringes; although this effect is small for the fringe ranges measured here, it would become non-negligible for very high orders or short screen distances.

7 Conclusion

This experiment measured the diameter of a human hair using laser diffraction and tested whether the Fraunhofer single-slit formula $y_m = m\lambda L/a$ gives consistent results across two wavelengths. Applying Babinet’s principle, the hair was treated as a single slit of the same

width. A forced-origin linear fit to the fringe position data yielded:

$$a_{\text{red}} = 85.1 \pm 1.3 \text{ }\mu\text{m} \quad (\lambda = 650 \text{ nm}, L = 100 \text{ cm}),$$

$$a_{\text{green}} = 89.0 \pm 1.7 \text{ }\mu\text{m} \quad (\lambda = 532 \text{ nm}, L = 200 \text{ cm}).$$

Both values lie within the established range for human hair (17–181 μm) [12], and both linear fits achieve $R^2 > 0.9999$, confirming that fringe position scales linearly with order number as wave optics predicts.

The two measurements differ by 3.9 μm , corresponding to a Z-score of 1.82, which falls inside the $|Z| < 1.96$ threshold for statistical consistency at 95% confidence. Their near-agreement confirms that the diffraction model works reliably across wavelengths, consistent with the findings of Ditchet et al. (2020) [10]. This shows that diffraction metrology relying on the Fraunhofer single-slit formula is a reliable measurement technique.

The dominant limitation of the experiment, however, is the laser wavelength uncertainty, which accounts for over 98% of the total error budget. Replacing the consumer laser with a calibrated spectrometer — or using a laser with a tightly specified wavelength — would reduce this uncertainty by an order of magnitude and substantially sharpen the statistical comparison between the two wavelengths.

References

- [1] D. Konstan, “Ancient Atomism,” *Stanford Encyclopedia of Philosophy*, Fall 2023 ed. <https://plato.stanford.edu/archives/fall2023/entries/atomism-ancient/>
- [2] T. Ainsworth, “Aristotle on Hylomorphism,” *PhilArchive*. <https://philarchive.org/archive/ANAAAH>
- [3] Author unknown, “Light in ancient Indian philosophy,” *arXiv:physics/0701077* (2007). <https://arxiv.org/pdf/physics/0701077>
- [4] “This Month in Physics History: May 1678 — Huygens proposes wave theory,” *APS News*, vol. 17, no. 5 (May 2008). <https://www.aps.org/archives/publications/apsnews/200805/physicshistory.cfm>
- [5] Imperial College London, “The double-slit experiment that proved the wave nature of light” (2022). <https://www.imperial.ac.uk/news/244037/double-slit-experiment-that-proved-wavenature/>
- [6] Author unknown, “Wave-particle duality,” *arXiv:1611.00226* (2016). <https://arxiv.org/pdf/1611.00226>
- [7] A. Duffy, “Huygens’ Principle,” Boston University Physics. https://physics.bu.edu/~duffy/semester2/d24_huygens.html
- [8] PhysLab, *Diffraction*, Ch. 10. <https://physlab.org/wp-content/uploads/2018/08/Diffraction-Ch10.pdf>
- [9] Physics Open Lab, “Light Diffraction and Babinet’s Principle” (2021). <https://physicsopenlab.org/2021/01/21/light-diffraction-and-babinet-principle/>
- [10] Ditchet et al., “Multi-wavelength diffraction metrology,” (*Journal name*), 2020. <https://www.tandfonline.com/doi/10.1080/00332747.2020.1768008>

- [11] “Reliability and aging of diode lasers,” *IEEE (Proceedings)*. <https://ieeexplore.ieee.org/document/311456>
- [12] BioNumbers, “Diameter of a human hair,” Harvard Medical School. <https://bionumbers.hms.harvard.edu/bionumber.aspx?id=106856&ver=2>
- [13] Harklinikken, “Hair density and thickness,” <https://harklinikken.com/blogs/journal/are-hair-density-and-thickness-the-same>